Tips for Solving Polynomials

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- rational root theorem to find candidates for roots of the polynomial
- polynomial remainder theorem to check if a polynomial is divisible
- synthetic division/ Horner's method to carry out the division

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Example: book 6th ed. 5.3/18

Diagonalise the following matrix:

$$\begin{bmatrix} -7 & -16 & 4 \\ 6 & 13 & -2 \\ 12 & 16 & 1 \end{bmatrix}$$

Example diagonalisation

Find the characteristic polynomial

First thing we have to do is find the characteristic polynomial $|A - \lambda I|$. I assume a working knowledge of determinants, so I leave it to the reader to verify that the polynomial is $-\lambda^3 + 7\lambda^2 + 5\lambda - 75$. Let's name it P.

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Then $q = \pm 1$ and we're looking only for integer roots.

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In the polynomial $P(\lambda) = -1 \cdot \lambda^3 + 7 \cdot \lambda^2 + 5 \cdot \lambda - 75$:

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- the free term is -75.

Therefore, if there are any rational roots, they must be of the form $\frac{p}{\pm 1}$ where p is a divisor of 75. That is, they are from the set $\pm \{1, 3, 5, 15, 25, 75\}$.

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Use the polynomial remainder theorem

It tells us that if $P(a) = 0$, then a is a root of P. We can try a candidate and see if the polynomial evaluates to 0.

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Try some candidate

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Use synthetic division

Instead of trying all other candidates, I recommend that you factor the polynomial into two terms. This approach is certain to deal with polynomials of any degree while guessing the candidates might fail.

Factoring the cubic

Use synthetic division to reduce the degree of the polynomial

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Factor the following polynomials

1. $x^3 - 2x^2 - 5x + 6$
2. $-x^3 + 5x^2 - 3x - 9$
3. $x^3 + 6x^2 + 12x + 8$
4. $x^3 + 5x^2 - 9x - 45$
5. $x^4 + x^3 - 15x^2 + 23x - 10$